

# Lab: Enactivism, 4E Cognition, and Active Inference

**Learning Goal:** Evaluate the philosophical tensions between Active Inference and 4E cognition.

## Part 1: 4E Audit

**Exercise:** For each of the 4 E's, evaluate Active Inference's engagement (1-5 scale):

| E        | Definition | How AI Engages | Score (1-5) | Key Tension |
|----------|------------|----------------|-------------|-------------|
| Embodied |            |                |             |             |
| Embedded |            |                |             |             |
| Enacted  |            |                |             |             |
| Extended |            |                |             |             |

Write a 200-word analysis: Which E does Active Inference handle best? Which worst?

*Write your response here*

## Part 2: The Representation Debate

**Learning Goal:** Clarify what's at stake in the enactivism vs. representationalism debate.

**Exercise:** Consider these three positions on what "generative model" means:

**A) Strong representationalism:** The generative model literally represents the world — it's a map of reality inside the brain.

**B) Moderate functionalism:** The generative model is a functional structure that covaries with environmental regularities. Whether it "represents" depends on your definition.

**C) Radical enactivism:** There is no generative model — there's only dynamic coupling between organism and environment. Talk of "models" is a computational metaphor imposed by theorists.

Argue for each position (150 words each), then defend your own view.

*Write your response here*

## Part 3: The Extended Mind Challenge

**Learning Goal:** Analyze whether Markov Blankets block or enable cognitive extension.

**Exercise:** Consider these scenarios:

1. A blind person navigating with a cane
2. A mathematician using a whiteboard
3. A group of people making a decision through deliberation
4. A person using GPS navigation

For each: (a) Where is the Markov Blanket? (b) Is cognition "extended" beyond the brain? (c) Can nested blankets capture the extension?

*Write your response here*

## Part 4: Cognitive Niche Construction Case Study

**Learning Goal:** Apply niche construction theory using Active Inference.

**Exercise:** Choose one of the following and analyze it as cognitive niche construction:

- The invention of writing
- The development of mathematics
- The construction of a library or university
- The creation of the internet

How does the innovation reduce free energy? How does it modify the cognitive niche? What new Markov Blankets does it create?

*Write your response here*

## Part 5: Reflection

In 250 words, reflect: Is Active Inference ultimately a representationalist framework in 4E clothing, or does it genuinely transcend the representationalism/enactivism divide? What would a truly post-representational Active Inference look like?

*Write your response here*

## Lab Summary

| Part | Skill Practiced             | Key Concept                  |
|------|-----------------------------|------------------------------|
| 1    | Framework evaluation        | 4E cognition audit           |
| 2    | Dialectical analysis        | Representation debate        |
| 3    | Applied analysis            | Extended mind + MB           |
| 4    | Case study analysis         | Cognitive niche construction |
| 5    | Meta-theoretical reflection | Beyond the divide            |